

Efficient numerical methods for the contact model

Local memory optimization and reusable contact topology

7 October 2026

Scope. This work preserves the declared fixed-mode spring/slider law and adds dimension-independent scheduling for complete force-and-angular-impulse contact blocks. It does not finish the general collision model, validate new material parameters, or establish superiority over another physics engine.

What was implemented

The contact-history solver prepares its small Cholesky factors and face metadata once. It uses the original full unconstrained arithmetic for every static-capacity decision. On yielding it can try a caller-owned previous face, then a face predicted by the current unconstrained solution. Every trial must satisfy the current box constraints and KKT conditions. If the guesses fail, the complete original face search remains available. Zero capacity is a singleton interval. Opening retains its separate released-mode-energy ledger; finite/passivity checks remain mandatory. No inverse is formed and no friction, restitution, stiffness or mass value is fitted.

Previous-face hints are not persistent physical state. Contact identity and the existing mode history remain the caller's responsibility. The prepared mobility, stiffness and timestep must remain fixed; changing them requires a new prepared object. The component still covers only one to three declared modes.

The C++17 indexed scheduler connects contacts through mutable bodies, splits islands, and builds conflict-free colors. A complete contact includes both its force impulse and independent angular impulse; the body angular update also includes the force lever moment. A shared read-only support does not merge independent bodies. Color search is bounded; excess conflicts run serially. Cache reuse checks every ordered endpoint, mutable-body flag and color budget. An invalid request cannot corrupt the previous plan.

The optional OpenMP visitor has barriers between colors. It currently commits prescribed or already gated blocks; it is not a qualified colored nonlinear world solver. Other joints and shared mutable constitutive states must be included before islands can be treated as independent solves.

Research basis and next step

Warm starts, substeps and constraint variants are described in [Catto's Solver2D](#); persistent connectivity in [Simulation Islands](#); and graph coloring/data layout in [Box2D 3.0](#). [Jolt's architecture](#) is a reference for multicore island processing. [MuJoCo's computation reference](#) includes coupled angular friction, but its material law is not a replacement for the requested motion-directed law.

Next priority is a body-local sparse/matrix-free application of the article's full shared-point wrench mobility. Accumulate signed force and couple momentum, apply inverse mass/full world inertia, gather relative motions and project into the allowed component directions. This preserves inter-contact coupling without a dense contact matrix. Qualify it against dense 2D/3D oracles before adoption.

Paired local-history timings

Seven alternating repetitions after warmup; same inputs and consumed outputs. The table reports medians for 10,000 repeated Python local responses, not an evolving world trajectory. Motion/history evolution is tested separately. Validation, triangular solves, current friction checks and energy gates are included. World discovery, integration, group solving and preparation are excluded. These timings do not replace the earlier native supported-response benchmark. Host: AMD Ryzen 7 3700X, Linux x86-64, Python 3.14.4, NumPy 2.5.3.

“Current” is the default no-hint solver; “Warm” also supplies the previous accepted face. The frozen implementation is retained byte-for-byte. The coherent lower case deliberately favors the original enumeration order; upper and mixed cases expose its ordering cost. Reversing cases invalidate old hints. Coupled-face cases test a wrong unconstrained guess and a valid previous face.

Modes	Case	Frozen ms	Current ms	Warm ms	Gain current/warm
1	Static/sticking	626.3	586.3	591.4	1.07/1.06
1	Coherent lower	1058.5	959.3	979.5	1.10/1.08
1	Reversing	1491.8	979.4	1242.5	1.52/1.20
1	Coherent upper	1875.7	960.8	980.8	1.95/1.91
1	Mixed directions	1920.8	977.8	992.8	1.96/1.93
2	Static/sticking	645.5	625.4	625.1	1.03/1.03
2	Coherent lower	1112.8	1009.9	1043.2	1.10/1.07
2	Reversing	2919.0	1001.4	1282.9	2.91/2.28
2	Coherent upper	4743.7	1018.1	1035.9	4.66/4.58
2	Mixed directions	3906.3	1023.9	1061.9	3.82/3.68
2	Coupled face	1708.9	1583.0	1097.8	1.08/1.56
3	Static/sticking	636.8	597.5	600.2	1.07/1.06
3	Coherent lower	1088.2	988.5	1012.7	1.10/1.07
3	Reversing	7213.2	1043.2	1293.8	6.91/5.58
3	Coherent upper	13368.6	1036.1	1055.3	12.90/12.67
3	Mixed directions	10416.5	1014.1	1028.2	10.27/10.13
3	Coupled face	3187.9	2431.8	1114.3	1.31/2.86

All 51 size/scenario combinations are retained in the machine-readable results (100, 1,000 and 10,000 updates). Across them the default gain is 1.03–12.97 times; with hints it is 1.02–12.67 times. These are measured local numerical gains, not an all-scene 2x gate or a competitor ranking. A hint can cost time when it fails or when the current prediction already succeeds.

Preparation is deliberately reported separately: frozen objects took 0.068–0.199 ms; prepared candidates took 0.106–0.842 ms. Candidate preparation includes all zero-capacity patterns and their small factor metadata. Frequent mobility changes can reduce or erase its amortized benefit.

The preliminary v2/v3/v4 runs and candidates are preserved but not accepted as final performance. An exact-static-limit audit found 9 branch changes from roundoff in the fast unbounded solve; differing static/dynamic friction could then cause large motion changes. The final solver retains the original unbounded trial for all static decisions, without changing any material threshold. The initial SLSQP oracle stopped short on one scaled quadratic; its failure log is retained. The final independent oracle uses an equivalent bounded least-squares transform and SciPy BVLS, eliminating zero-capacity coordinates explicitly.

Topology costs and correctness

One million contacts per graph; seven alternating fresh builds/cache hits. Single-worker C++17 -O3, without fast-math. Fresh-build timing includes planner allocation/construction, stops before plan destruction, and excludes input graph creation. Reuse includes full equality checking and lookup; no allocation on a hit. These costs exclude collision discovery and all physical solving.

Graph	Islands	Colors	Serial tail	Build ms	Reuse ms
independent_floor	1,000,000	1	0	27.84	1.00
chain	1	2	0	24.61	0.95
hub	1	32	999,968	14.97	0.85
heterogeneous_degree	41,667	7	0	22.84	0.87

The million-contact hub illustrates the ownership limit: nearly every contact shares one dynamic body, so coloring cannot manufacture parallel independence. The heterogeneous-degree case is graph topology, not simulated irregular-shape geometry. Shapes and materials do not enter this planner.

Verification. All 63 focused repository tests pass, including the existing supported, restitution, measured-inertia and conservation checks. 1500 seeded local-history cases (910 yielded) agree with the frozen physical outputs to 2.17e-19 maximum scaled difference. 141 cases were independently checked by BVLS; the maximum impulse difference was 5.55e-17. Sequential motion/history, opening/recontact, zero capacities, singular body mobility, wrong face hints and coupled fallback have explicit controls. An additional 300 exact/adjacent-static-limit controls preserve every original static/dynamic branch; static impulses are bit-identical to the original.

The native planner passes 300 randomized graph comparisons with an independent BFS connectivity reference and write-conflict checks. Prescribed 2D/3D force plus independent-couple applications verify linear and angular momentum, including prescribed-support reactions. Eight-worker and serial colored outputs are bit-identical. Both OpenMP and portable serial builds pass. These are mechanics identities, not restitution/friction predictions of measured impacts.

Remaining limits. General moving full t/s history, zero-motion directional closure, joint normal impact, a physical coupled finite-patch budget, and dense irregular/group trajectory accuracy remain open. Coloring can change nonlinear iteration convergence and must be qualified on complete scenes. Existing public material parameters and experimental errors are unchanged; no empirical accuracy gain is claimed. The main compiler and native world defaults are unchanged, as is the supported-response C ABI v1. This planner is a C++ header interface, not a completed compiler/GPU port.

Reproducibility. Final code/harness hashes are in run-v5/results.json; earlier checkpoints e66c4d5 and f7e5b8f retain preliminary versions. Frozen source, earlier outcomes, final sample timings, source hashes, independent audits and tests are retained in [research/modern-contact-optimization](#). The separate symbolic article is preserved.